Lec_53

Practice Question on Heap

52

Lec-53

Algo grade Exam

oct

A to Z

Lec-0

80%

Greedy

dp

pyq

51, 52

1.)What is worst case TC

$\rightarrow O(1)$ always

$\rightarrow O(n/2) = O(n)$

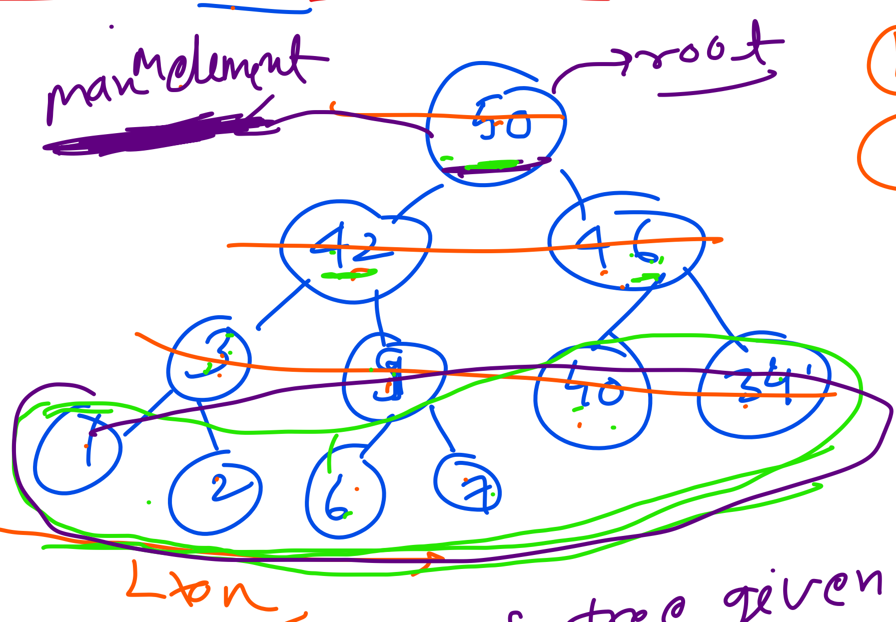
$\rightarrow O(1)$

- (a) **To find Maximum and Minimum Element in Max heap**
- (b) **To find Maximum and Minimum Element in Min heap**

or CBT

$= O(n) = O(n/2)$

(a) Max Heap



(1) ACBT ✓

(2) To get maxm element in max heap simply return root→data

→ O(1)

→ Tree given means, root of tree given

in max heap root→data is maximum element

→ In max-Heap minimum element is present in leaf nodes

→ # If a Binary tree has n nodes,

Internal nodes = $\lceil n/2 \rceil$

leaf nodes = $\lfloor n/2 \rfloor$

$\lceil 4.5 \rceil = 5$

$\lfloor 4.5 \rfloor = 4$

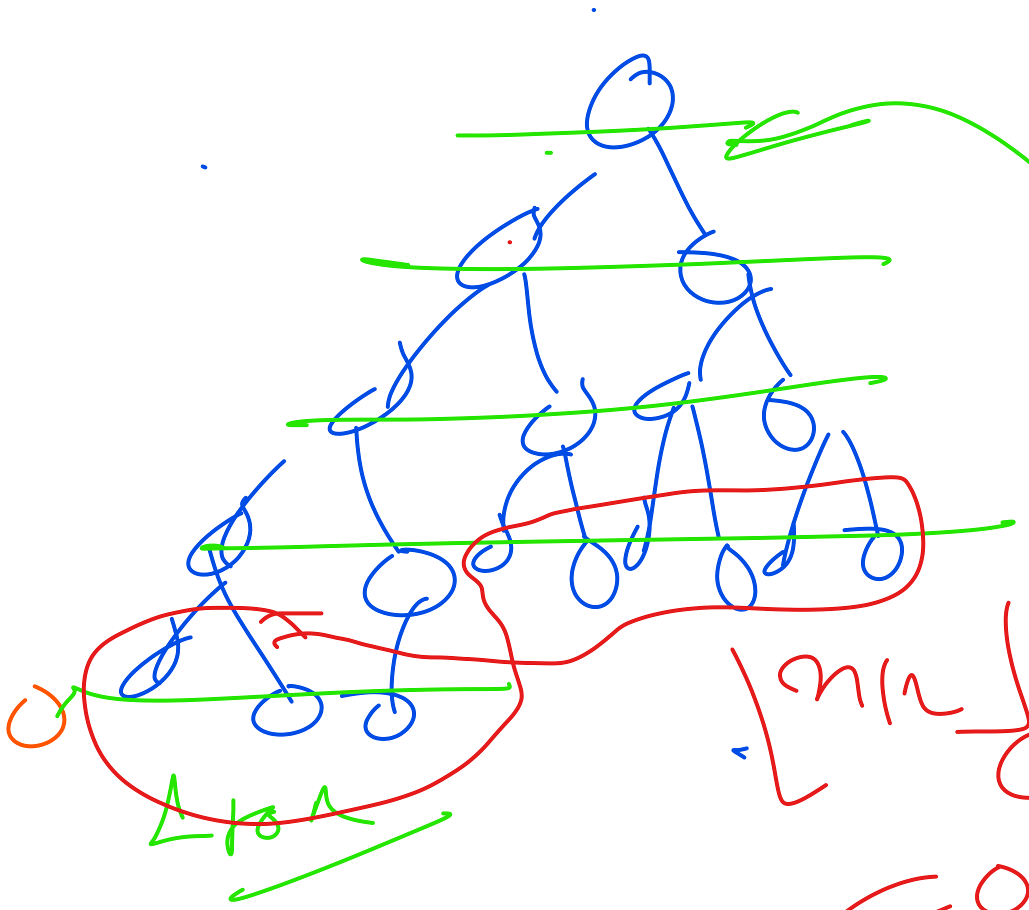
→ And we know that minimum element in max heap is at leaf nodes, and we know that $\lfloor n/2 \rfloor$ elements are there in max heap.

In $n/2$ elements to find minimum element

$n/2$

Time complexity $\Rightarrow O(\frac{n}{2})$

$= O(n)$



ACBT

max

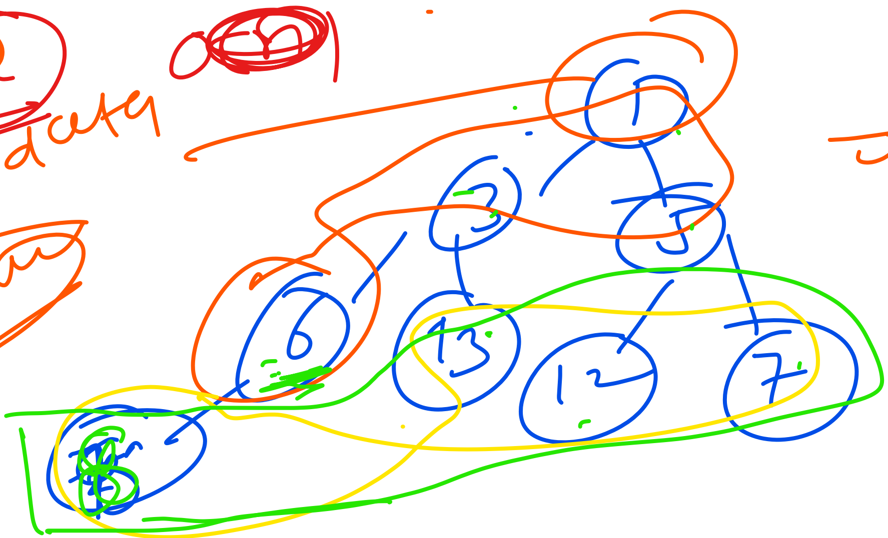
$n/2$

$\lceil n/2 \rceil \Rightarrow O(n/2)$

$= O(n)$

root + data

Minimum



$\Rightarrow$ min heap (root given)

min $= O(1)$

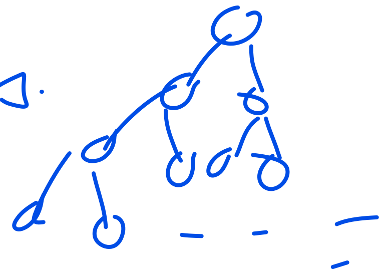
for max^m in $n/2$

$n/2 \Rightarrow$ elements $= O(n/2)$

$= O(n)$

2.)In a Max Heap Tree 1023 Nodes are present ,to find minimum element how many comparisons?

→ In max Heap min elements is present at leaf nodes.



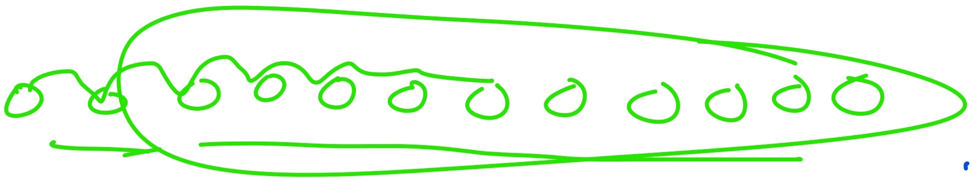
→ in n nodes BT , $\lfloor n/2 \rfloor$ leaf nodes are there

$n = 1023$

leaf nodes = $\lfloor 1023/2 \rfloor = \lfloor 511.5 \rfloor$

$= 511 \Rightarrow 510$

$= 510$



→ In n elements to find minimum How many we

$= n-1$,

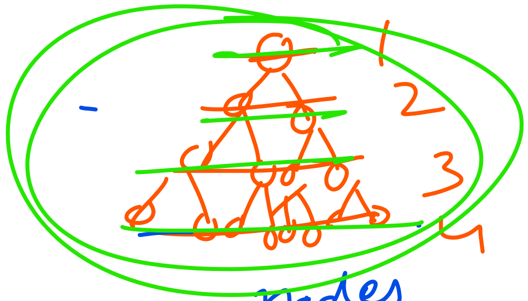
3.If in max Heap nodes are 1020 ,how many Comparisons for Min Element

Comment

4.)If number of levels =10,than

in CBT max = min = 1023

- Find max and min nodes in CBT
- Find max and min nodes in ACBT



	nodes
K=1	1
K=2	1+2
K=3	1+2+4
K=4	1+2+4+8

K=3, $2^0+2^1+2^2$

K=4, $2^0+2^1+2^2+2^3$

K=10, $2^0+2^1+2^2+2^3 - - - 2^9$

$2^0+2^1+2^2+2^3 - - - 2^9$

$\Rightarrow \frac{a(r^n-1)}{r-1} \Rightarrow \frac{1(2^{10}-1)}{2-1} \Rightarrow 2^{10}-1 \Rightarrow 1024-1 = 1023$

2^n-1

1+levels

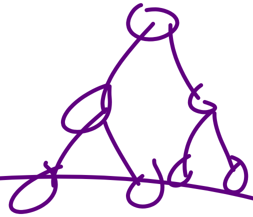
$\Rightarrow 2^{10}-1$

ACBT

ACBT

ACBT

CBT



→ every CBT is ACBT

maxm

level = 10

maxm = 1023

k=4

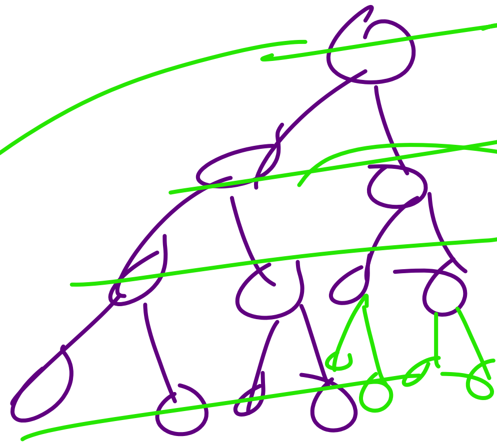
k=1

2

3

4

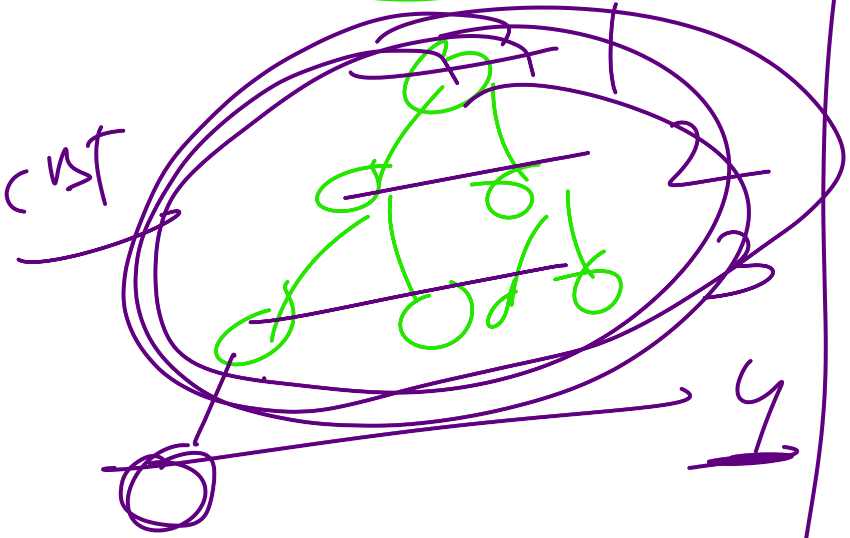
CBT

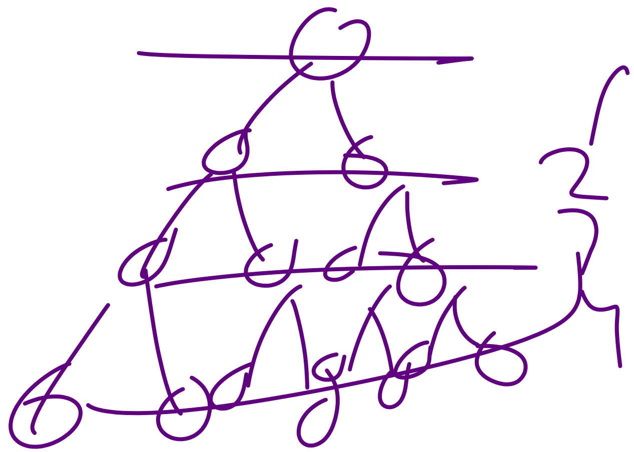


ACBT

k=4

CBT





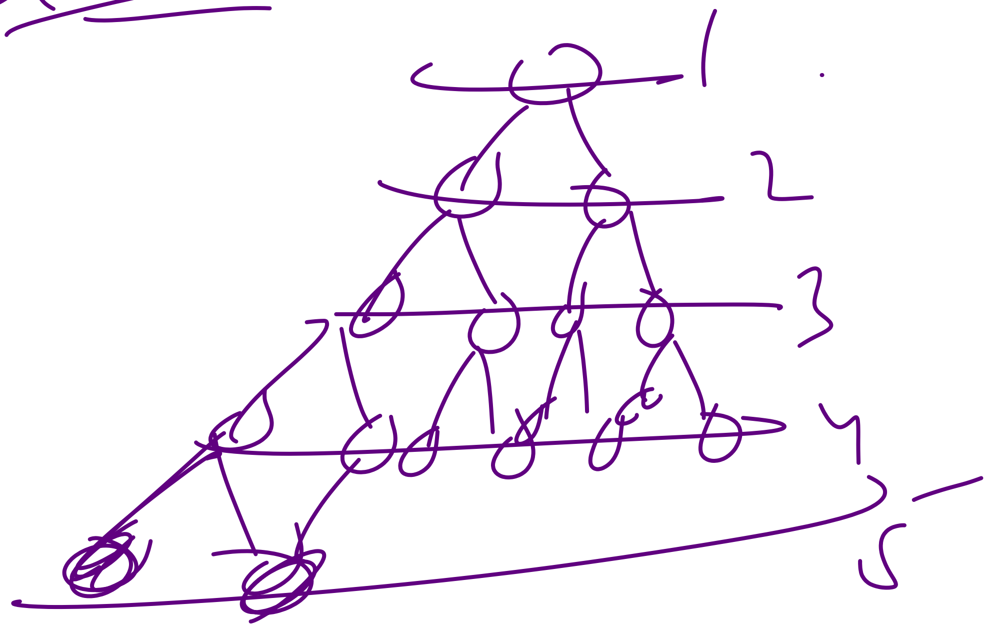
L to R

If levels are four than minimum count = CBT of levels + 1 if level is 10 → no of nodes in CBST of level 9 + 1

→ 1 $2^9 - 1 + 1$

→ 2^9 → ~~512~~

level 5



~~In CBST~~